

Landing to Bronze

Batch

```
python batch_landing_to_bronze.py --batch-date 2026-06-15 --tables  
customers_orgs,users,billing_monthly
```

Primera corrida:

```
[OK] customers_orgs: read=80 after_quality=80 after_dedupe=80 written=80  
[OK] users: read=800 after_quality=800 after_dedupe=800 written=800  
[OK] billing_monthly: read=240 after_quality=240 after_dedupe=240 written=240  
[VERIFY] bronze/customers_orgs total_rows=80 (use this value to compare reruns for  
idempotency)  
[VERIFY] bronze/users total_rows=800 (use this value to compare reruns for idempotency)  
[VERIFY] bronze/billing_monthly total_rows=240 (use this value to compare reruns for  
idempotency)  
[INFO] Manifest: datalake/bronze/_control/batch_date=2026-06-15/manifest.json  
[INFO] Totals: read=1120 after_quality=1120 after_dedupe=1120 written=1120
```

Segunda corrida:

```
[OK] customers_orgs: read=80 after_quality=80 after_dedupe=80 written=80  
[OK] users: read=800 after_quality=800 after_dedupe=800 written=800  
[OK] billing_monthly: read=240 after_quality=240 after_dedupe=240 written=240  
[VERIFY] bronze/customers_orgs total_rows=80 (use this value to compare reruns for  
idempotency)  
[VERIFY] bronze/users total_rows=800 (use this value to compare reruns for idempotency)  
[VERIFY] bronze/billing_monthly total_rows=240 (use this value to compare reruns for  
idempotency)  
[INFO] Manifest: datalake/bronze/_control/batch_date=2026-06-15/manifest.json  
[INFO] Totals: read=1120 after_quality=1120 after_dedupe=1120 written=1120
```

Particiones:

```
▼ bronze  
  > _control \ batch_date=2026-06-15  
  ▼ batch  
    ▼ billing_monthly  
      > month=2025-06-01  
      > month=2025-07-01  
      > month=2025-08-01  
    > customers_orgs \ load_date=2026-06-15  
    > users \ load_date=2026-06-15
```

Path	Size_KB
----	-----
batch\billing_monthly\month=2025-06-01\part-00000-3aa7888d-c7bb-4fa9-a3fb-36f580a97fa3.c000.snappy.parquet	7.18
batch\billing_monthly\month=2025-07-01\part-00000-3aa7888d-c7bb-4fa9-a3fb-36f580a97fa3.c000.snappy.parquet	7.16
batch\billing_monthly\month=2025-08-01\part-00000-3aa7888d-c7bb-4fa9-a3fb-36f580a97fa3.c000.snappy.parquet	7.16
batch\customers_orgs\load_date=2026-06-15\part-00000-90bd3ef3-3688-4152-ad60-6defb5df1ea2.c000.snappy.parquet	7.16
batch\users\load_date=2026-06-15\part-00000-ed443038-c381-4837-b7f6-e79cca73fa72.c000.snappy.parquet	25.02

Streaming

python streaming_landing_to_bronze.py --watermark-delay "2 days" --max-files-per-trigger 50

Primera corrida:

[VERIFY] bronze/streaming/usage_events total_rows=36255 (compare this value across reruns for idempotency)

[VERIFY] bronze/quarantine/usage_events_invalid total_rows=0

Segunda corrida:

[VERIFY] bronze/streaming/usage_events total_rows=36255 (compare this value across reruns for idempotency)

[VERIFY] bronze/quarantine/usage_events_invalid total_rows=0

Particiones:

Path	Size_KB
----	-----
usage_events\event_date=2025-07-03\part-00000-14e4d188-5ce4-45ca-ab42-39a6531e2d8e.c000.snappy.parquet	11.65
usage_events\event_date=2025-07-03\part-00000-5508d150-290e-496b-87e8-fb24a57f89bd.c000.snappy.parquet	10.69
usage_events\event_date=2025-07-03\part-00001-55c55bf5-4cbf-42b8-925f-2c16a653769d.c000.snappy.parquet	12.65
usage_events\event_date=2025-07-03\part-00001-a1fb8568-c8d7-4f78-9fe2-3eb1091d13f5.c000.snappy.parquet	11.87
usage_events\event_date=2025-07-03\part-00002-aaf6c6a8-36f1-4411-b67f-e86a45f2f288.c000.snappy.parquet	11.83
usage_events\event_date=2025-07-03\part-00002-c0f45212-a110-45d0-ab33-ace908bceb9e.c000.snappy.parquet	11.1
usage_events\event_date=2025-07-03\part-00003-320c29e7-6e44-4ba9-b00c-8a8b14997641.c000.snappy.parquet	10.05
usage_events\event_date=2025-07-03\part-00003-d599c228-c684-4806-8d9b-1a65de3aec01.c000.snappy.parquet	12.81
usage_events\event_date=2025-07-04\part-00000-c3c4215e-1600-4c9a-a26a-1e48ac70eb27.c000.snappy.parquet	11.29
usage_events\event_date=2025-07-04\part-00000-e560c3ef-c633-4116-b393-3ff69912d0a6.c000.snappy.parquet	11.99
usage_events\event_date=2025-07-04\part-00001-c00a4eee-8140-4f8b-a51b-f1ab2be4e9d0.c000.snappy.parquet	11.97
usage_events\event_date=2025-07-04\part-00001-c9db9d76-7418-40ac-90e8-bb695d4ef499.c000.snappy.parquet	11.9
usage_events\event_date=2025-07-04\part-00002-1795bf35-b930-4e0f-8eef-c0d1d2ebd7fb.c000.snappy.parquet	11.36
usage_events\event_date=2025-07-04\part-00002-b2c5cec4-8929-431d-ad13-a6059e3fcbdd.c000.snappy.parquet	12.76
usage_events\event_date=2025-07-04\part-00003-20f28a4b-c795-4c14-a1f2-16a3b68445bf.c000.snappy.parquet	11.71
usage_events\event_date=2025-07-04\part-00003-5c6d8970-2922-4647-aa91-0557685ba1b8.c000.snappy.parquet	10.59
usage_events\event_date=2025-07-05\part-00000-9c4a34f0-24d8-4ab1-8eab-235de2976f6b.c000.snappy.parquet	12.49
usage_events\event_date=2025-07-05\part-00000-d75e56f0-b0dc-46ce-8bb9-ba12e189580d.c000.snappy.parquet	11.68
usage_events\event_date=2025-07-05\part-00001-a4a8b588-66b1-4651-bfe8-f774d80c8fb5.c000.snappy.parquet	11.22
usage_events\event_date=2025-07-05\part-00001-c81e318b-c20b-4e77-ad10-6afe02f7fed9.c000.snappy.parquet	10.96
usage_events\event_date=2025-07-05\part-00002-373490ba-912e-45dc-809b-7a7ea406eae0.c000.snappy.parquet	12.33
usage_events\event_date=2025-07-05\part-00002-b27d6d59-f491-4c46-a29c-f49fdec29412.c000.snappy.parquet	11.61
usage_events\event_date=2025-07-05\part-00003-46c7f56d-4df3-42c1-94e8-870a0ffd0131.c000.snappy.parquet	12.44
usage_events\event_date=2025-07-05\part-00003-52070b49-2de5-43df-bf10-96bf1169c3e0.c000.snappy.parquet	11.49

```
▼ streaming\usage_events
  > _spark_metadata
  > event_date=2025-07-03
  > event_date=2025-07-04
  > event_date=2025-07-05
  > event_date=2025-07-06
  > event_date=2025-07-07
  > event_date=2025-07-08
  > event_date=2025-07-09
  > event_date=2025-07-10
  > event_date=2025-07-11
  > event_date=2025-07-12
  > event_date=2025-07-13
  > event_date=2025-07-14
  > event_date=2025-07-15
  > event_date=2025-07-16
  > event_date=2025-07-17
  > event_date=2025-07-18
  > event_date=2025-07-19
  > event_date=2025-07-20
  > event_date=2025-07-21
  > event_date=2025-07-22
  > event_date=2025-07-23
```

Bronze to Silver

```
python streaming_landing_to_bronze.py --watermark-delay "2 days" --max-files-per-trigger 50
```

Primera corrida:

[VERIFY] silver/events_enriched total_rows=34546 (compare this value across reruns for idempotency)

[VERIFY] silver/features_org_daily total_rows=10439 (compare this value across reruns for idempotency)

[VERIFY] silver/quarantine/events_quality_issues total_rows=1709 (compare this value across reruns for idempotency)

[INFO] Bronze records: 36255

[INFO] After dedupe: 34546

[INFO] Quarantine records: 1709

[INFO] Silver events: 34546

[INFO] Silver features: 10439

Segunda corrida:

[VERIFY] silver/events_enriched total_rows=34546 (compare this value across reruns for idempotency)

[VERIFY] silver/features_org_daily total_rows=10439 (compare this value across reruns for idempotency)

[VERIFY] silver/quarantine/events_quality_issues total_rows=1709 (compare this value across reruns for idempotency)

[INFO] Bronze records: 36255

[INFO] After dedupe: 34546

[INFO] Quarantine records: 1709

[INFO] Silver events: 34546

[INFO] Silver features: 10439

[OK] Silver pipeline completed. Manifest: datalake/silver/_control/manifest.json

Silver to Gold

```
python silver_to_gold.py --max-files-per-trigger 1000
```

Primera corrida:

[INFO] Silver input records: 10439

[INFO] Gold output records: 10439

Segunda corrida:

[INFO] Silver input records: 10439

[INFO] Gold output records: 10439

[VERIFY] gold/org_daily_usage_by_service total_rows=10439 (compare this value across reruns for idempotency)